CHAPTER 1 | LOGIC AND PROOFS

Discrete Mathematics • Challenge Test • Worked Solutions

Marking principle. The total is 100 marks. Award credit for mathematically equivalent answers and valid alternative proofs unless a particular method is required. The allocations below are a suggested marking scheme.

1. System requirements [8 marks]

(a) [4] One mark for each correct translation:
$R1: r \to (p \vee q)$
$R2: q \to \neg p$
$R3: (p \wedge \neg q) \to r$
$R4: q \to r$

(b) [2] The complete set is $\{(F,F,F), (T,F,T), (F,T,T)\}$.
If $q=T$, R2 forces $p=F$ and R4 forces $r=T$. If $q=F$ and $p=T$, R3 forces $r=T$. If $p=q=F$, R1 forces $r=F$. These cases exhaust all possibilities, and each listed assignment satisfies all four requirements.

(c) [2] Yes. R2 prevents p and q from both being true. R3 and R4 ensure admission whenever either is true; R1 prevents admission when both are false. Thus admission occurs exactly when one of p,q is true: $r \leftrightarrow (p \oplus q)$.
Common trap: "only if" in R1 gives $r \to (p \vee q)$, not the reverse.

2. Truth table [8 marks]

p	q	r	$p \oplus q$	A	$p \to r$	$q \to r$	B	F
T	T	T	F	T	T	T	T	T
T	T	F	F	T	F	F	F	F
T	F	T	T	T	T	T	T	T
T	F	F	T	F	F	T	F	T
F	T	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F	T
F	F	T	F	T	T	T	T	T
F	F	F	F	T	T	T	T	T

(a) [6] Suggested split: $p \oplus q$ and A, 2 marks; the two implication columns and B, 2 marks; the final column, 2 marks.

(b) [1] F is a **contingency**: it is true in seven rows and false in one.

(c) [1] $F \equiv \neg p \vee \neg q \vee r$. Its only false assignment is $p=q=T, r=F$. In that row the XOR antecedent is false, so $A=T$, but both implications in B are false. The formula is also equivalent to $(p \wedge q) \to r$, though the requested connective restriction requires the disjunctive form.

3. Equivalence and NOR [10 marks]

(a) [6] Let $L=(p\wedge q)\vee(\neg p\wedge r)\vee(q\wedge r)$. Insert the tautology $p\vee\neg p$ into the last term:

$L \equiv (p\wedge q) \vee (\neg p\wedge r) \vee [(q\wedge r)\wedge(p\vee\neg p)]$
Identity and excluded middle.
$\equiv (p\wedge q) \vee (\neg p\wedge r) \vee (p\wedge q\wedge r) \vee (\neg p\wedge q\wedge r)$
Distributivity, associativity, and commutativity.
$\equiv [(p\wedge q) \vee ((p\wedge q)\wedge r)] \vee [(\neg p\wedge r) \vee ((\neg p\wedge r)\wedge q)]$
Associativity and commutativity.
$\equiv (p\wedge q) \vee (\neg p\wedge r)$
Absorption applied twice. The term $q\wedge r$ is redundant.

Marks: inserting $p\vee\neg p$, 2; distributing and regrouping, 2; both absorptions and conclusion, 2.

(b) [4] One valid construction is
$[p \downarrow (p \downarrow q)] \downarrow [q \downarrow (p \downarrow q)]$.
Let $u=p\downarrow q=\neg p\wedge\neg q$. Then
$p\downarrow u = \neg(p\vee(\neg p\wedge\neg q)) = \neg(p\vee\neg q) = \neg p\wedge q$,
$q\downarrow u = \neg(q\vee(\neg p\wedge\neg q)) = \neg(q\vee\neg p) = p\wedge\neg q$.
The final NOR is therefore
$\neg[(\neg p\wedge q)\vee(p\wedge\neg q)] \equiv p\leftrightarrow q$.
It is true exactly when the two input truth values agree.
Marks: a NOR-only construction, 2; correct verification, 2.

4. Order and domain [8 marks]

(a) [2] True over $\mathbb{Z}$. For each integer x, choose $y=-x$. Then $x+y=0\le z^2$ for every integer z. The witness is chosen before z and does not depend on it.

(b) [2] False over $\mathbb{Z}$. For any proposed integer y, choose $x=-y$. The existential requirement becomes $z^2<0$, which no integer satisfies. One failed candidate for y would not suffice; this argument defeats **every** candidate.

(c) [2] False over $\mathbb{Z}$. Take $x=0$. The requirement $0<2y<2$ gives $0<y<1$, and there is no integer in this interval.

(d) [2] True over $\mathbb{R}$. Given any real x, choose $y=(x+1)/2$. Then $2y=x+1$, so $x<2y<x+2$. The integer-domain obstruction disappears because this witness is always a real number.

5. Restricted uniqueness [12 marks]

(a) [4]

$\forall y \exists x[x \neq y \wedge R(x,y) \wedge \forall z((z \neq y \wedge R(z,y)) \rightarrow z=x)]$.

For each report owner y, the witness x is a different student who reviews the report. Every other eligible reviewer must equal x. The formula permits $R(y,y)$: self-review was not forbidden by the English sentence.

Marks: quantifier order and existence, 1; exclusion of self, 1; uniqueness restricted to other students, 2.

(b) [4] For explanation only, write

$U(x) = \neg R(x,x) \wedge \forall y(y \neq x \rightarrow R(x,y))$.

The required final formula, with that abbreviation expanded, is:

$$\exists x[\neg R(x,x) \wedge \forall y(y \neq x \rightarrow R(x,y)) \wedge \forall z((\neg R(z,z) \wedge \forall w(w \neq z \rightarrow R(z,w))) \rightarrow z=x)].$$

The last clause excludes any second student having the entire stated property. It does not prohibit other students from reviewing some reports.

Marks: existence and the full property, 2; uniqueness of that property, 2.

(c) [4] Negating (a) and moving the negation inward gives

$\exists y \forall x[x=y \vee \neg R(x,y) \vee \exists z(z \neq y \wedge R(z,y) \wedge z \neq x)]$.

In detail, the outer $\forall y \exists x$ becomes $\exists y \forall x$. The negation of the uniqueness clause is $\exists z(z \neq y \wedge R(z,y) \wedge z \neq x)$. The final formula contains no implication and no negation of a compound formula.

English: Some student has either no reviewer other than themselves, or at least two distinct reviewers other than themselves.

Marks: correct outer quantifiers, 1; correct three-way disjunction and witness condition, 2; English meaning, 1.

6. Finite expansion and two failed implications [8 marks]

(a) [3]

$(R(a,b) \vee R(a,c)) \wedge (R(b,a) \vee R(b,c)) \wedge (R(c,a) \vee R(c,b))$.

The self-pairs disappear because $x \neq x$ is false. Award one mark for each correct conjunct.

(b) [2] A does not imply B. Let

$R=\{(a,b),(b,c),(c,a)\}$.

Every element relates to a different element, so A is true. No possible common target satisfies B: target a fails at $x=b$, target b fails at $x=c$, and target c fails at $x=a$.

(c) [3] B does not imply A. Let

$R=\{(a,c),(b,c)\}$.

Choose $y=c$. Both other elements relate to c, and the implication for $x=c$ is vacuously true. Thus B is true. But c has no outgoing pair to a different element, so A is false.

In both models, every unlisted pair, including every self-pair, is false.

7. A justified derivation [10 marks]

Step	Statement	Reason
1	$E(a) \land \neg P(a)$	EI from IV; a fresh
2	$E(a)$ and $\neg P(a)$	Simplification, 1
3	$E(a) \to \exists y(M(y,a) \land T(y))$	UI from I
4	$\exists y(M(y,a) \land T(y))$	MP, 2 and 3
5	$M(b,a) \land T(b)$	EI, 4; b fresh
6	$(M(b,a) \land T(b)) \to A(a)$	UI from II
7	$A(a)$	MP, 5 and 6
8	$A(a) \to (P(a) \lor Q(a))$	UI from III
9	$P(a) \lor Q(a)$	MP, 7 and 8
10	$Q(a)$	DS, 9 and ¬P(a) from 2
11	$A(a) \land Q(a)$	Conjunction, 7 and 10
12	$\exists x(A(x) \land Q(x))$	EG, 11

Marks: first witness and extracted facts, 2; second witness, 2; deriving $A(a)$, 2; deriving $Q(a)$, 2; conjunction and EG, 2. The witnesses need not be different. The final conclusion contains neither witness name.

8. Proof audit and minimality [12 marks]

(a) [6] There are three independent errors, worth 2 marks each.
Line 6: b was chosen after a and may depend on a. It cannot be held fixed while a is universally generalized.
Line 9: b is an existential witness, not an arbitrary element. $P(b)$ does not establish $\forall x P(x)$.
Line 10: EI cannot assign the witness of premise III to the already used b. A fresh name is needed; the premise does not imply $\neg P(b)$.

Line 8 is valid within the witness reasoning. Lines 7 and 11 have valid rule forms, but depend on unsupported statements; they are not additional independent rule errors.

(b) [3] Take $D=\{0,1,2\}$, $P=\{0,1\}$, and
$R=\{(0,0),(1,1),(2,0)\}$, with all other pairs false.
Every element has an R-successor, so I holds. Every successor satisfies P, so II holds. Element 2 fails P, so III holds. No element is a common successor: target 0 fails at x=1; target 1 fails at x=0; and no pair ends in 2. Hence line 7 is false.

(c) [3] I and II require at least one P-element, while III requires a non-P-element. A one-element model is impossible. In a two-element model satisfying the premises, exactly one element, say u, satisfies P. II forces every R-successor to be u, and I requires every element to have a successor. Thus $\forall x R(x,u)$, making line 7 true. At least three elements are needed, and (b) supplies three.

9. Parity and impossibility [14 marks]

(a) [7] Forward direction, by contraposition. Suppose m and n are not both even. There are exactly three parity cases:

m	**n**	m^2	mn	n^2	**H(m,n)**
Odd	Even	Odd	Even	Even	Odd
Even	Odd	Even	Even	Odd	Odd
Odd	Odd	Odd	Odd	Odd	Odd

In every case H is odd, proving the contrapositive. **Reverse direction, directly:** if $m=2a$ and $n=2b$ for integers a,b, then
$H(m,n)=4(a^2+ab+b^2)$,
which is even. Hence H is even if and only if both inputs are even.
Marks: correct contrapositive, 1; all three cases, 3; direct reverse direction, 2; both-direction conclusion, 1.

(b) [5] Suppose integers m,n,k satisfy $H(m,n)=4k+2$. The right-hand side is even, so (a) gives $m=2a,n=2b$ for some integers a,b. Substitution gives
$4(a^2+ab+b^2)=4k+2$,
hence $2(a^2+ab+b^2-k)=1$. The left side is even and the right side is odd, a contradiction. Therefore no such integers exist.
Marks: contradiction assumption and use of (a), 2; substitution, 1; explicit parity contradiction and conclusion, 2.

(c) [2] Take $m=n=1$. Then $H(1,1)=3$ is odd although both inputs are odd. The correct equivalence is:
H(m,n) is odd if and only if at least one of m,n is odd.

10. Existence, uniqueness, and contradiction [10 marks]

(a) [6] Existence: Every integer n, positive, zero, or negative, is either even or odd. If $n=2k$, choose $(q,r)=(k,0)$. If $n=2k+1$, choose $(q,r)=(k,1)$. Both choices satisfy the required conditions.

Uniqueness: Suppose $n=2q+r=2q'+r'$, where both remainders lie in $\{0,1\}$. Then
$2(q-q')=r'-r$.
The left side is even; the right side lies in $\{-1,0,1\}$. Its only even possibility is 0. Thus $r'=r$, and then $q=q'$. The ordered pair is unique. No step assumed $n\geq 0$.
Marks: existence in both parity cases, 3; uniqueness of both coordinates, 3.

(b) [4] Fix a nonzero irrational real t. Assume, for contradiction, that both $u=t+1/t$ and $v=t-1/t$ are rational. Their sum $u+v=2t$ is rational. Dividing by the nonzero rational number 2 shows that $t=(u+v)/2$ is rational, contradicting the hypothesis. Therefore at least one of the two expressions is irrational.
Marks: correctly negate "at least one," 1; add the expressions, 1; rational closure, 1; contradiction and conclusion, 1.